

Assignment_5_AF

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```
library(RColorBrewer)
```

```
## Warning: package 'RColorBrewer' was built under R version 4.5.2
```

```
library(gamma)
```

```
## Warning: package 'gamma' was built under R version 4.5.2
```

```
library(DESeq2)
```

```
## Loading required package: S4Vectors
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
## Warning: package 'BiocGenerics' was built under R version 4.5.2
```

```
## Loading required package: generics
```

```
##
```

```
## Attaching package: 'generics'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
```

```
##      setequal, union
```

```
##
```

```
## Attaching package: 'BiocGenerics'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      IQR, mad, sd, var, xtabs
```

```

## The following objects are masked from 'package:base':
##
##   anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##   colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##   get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##   order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##   rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##   unsplit, which.max, which.min

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:utils':
##
##   findMatches

## The following objects are masked from 'package:base':
##
##   expand.grid, I, unname

## Loading required package: IRanges

##
## Attaching package: 'IRanges'

## The following object is masked from 'package:grDevices':
##
##   windows

## Loading required package: GenomicRanges

## Loading required package: GenomeInfoDb

## Loading required package: SummarizedExperiment

## Loading required package: MatrixGenerics

## Loading required package: matrixStats

##
## Attaching package: 'MatrixGenerics'

## The following objects are masked from 'package:matrixStats':
##
##   colAlls, colAnyNAs, colAnys, colAveragesPerRowSet, colCollapse,
##   colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##   colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##   colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##   colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##   colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##   colWeightedMeans, colWeightedMedians, colWeightedSds,

```

```
## colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
## rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
## rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
## rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
## rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
## rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
## rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
## rowWeightedSds, rowWeightedVars
```

```
## Loading required package: Biobase
```

```
## Welcome to Bioconductor
```

```
##
```

```
## Vignettes contain introductory material; view with
## 'browseVignettes()'. To cite Bioconductor, see
## 'citation("Biobase")', and for packages 'citation("pkgname")'.
```

```
##
```

```
## Attaching package: 'Biobase'
```

```
## The following object is masked from 'package:MatrixGenerics':
```

```
##
```

```
## rowMedians
```

```
## The following objects are masked from 'package:matrixStats':
```

```
##
```

```
## anyMissing, rowMedians
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:Biobase':
```

```
##
```

```
## combine
```

```
## The following object is masked from 'package:matrixStats':
```

```
##
```

```
## count
```

```
## The following objects are masked from 'package:GenomicRanges':
```

```
##
```

```
## intersect, setdiff, union
```

```
## The following object is masked from 'package:GenomeInfoDb':
```

```
##
```

```
## intersect
```

```
## The following objects are masked from 'package:IRanges':
##
## collapse, desc, intersect, setdiff, slice, union

## The following objects are masked from 'package:S4Vectors':
##
## first, intersect, rename, setdiff, setequal, union

## The following objects are masked from 'package:BiocGenerics':
##
## combine, intersect, setdiff, setequal, union

## The following object is masked from 'package:generics':
##
## explain

## The following object is masked from 'package:gamma':
##
## summarise

## The following objects are masked from 'package:stats':
##
## filter, lag

## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
```

```
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.5.2

##
## Attaching package: 'tidyr'

## The following object is masked from 'package:S4Vectors':
##
## expand
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(pheatmap)
```

```
## Warning: package 'pheatmap' was built under R version 4.5.2
```

```
library(RColorBrewer)
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.5.2
```

```
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.5.2
```

```
##
## Attaching package: 'kableExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##      group_rows
```

```
library(org.Hs.eg.db)
```

```
## Loading required package: AnnotationDbi
```

```
##
## Attaching package: 'AnnotationDbi'
```

```
## The following object is masked from 'package:dplyr':
##
##      select
```

```
##
```

```
# setwd("C:/Users/Delwar/Desktop/Humber/WINTER SEMESTER/BINF-5505 Applied Clinical Genetics/Assignments")
```

```
#Import Seq Data
```

```
#Load count data and annotation metadata
# cts <- read.csv("GSE205748_raw_counts_GRCh38.p13_NCBI.tsv", sep = "\t", row.names="GeneID")
# anot <- read.csv("dataset_annotation.csv")
# rownames(anot)<-anot$file

#Load count data and annotation metadata
cts <- read.csv("GSE205748_read_counts.csv", sep = "\t", row.names="ID")
anot <- read.csv("dataset_annotation.csv")
rownames(anot)<-anot$patient

# Match the ID column of df1 to the column names of df2
anot <- anot %>% arrange(match(rownames(anot), colnames(cts)))

print(rownames(anot))
```

```
## [1] "HC_6"      "HC_7"      "HC_8"      "HC_9"      "HC_11"     "HC_12"
## [7] "HC_13"     "HC_14"     "HC_15"     "PSA_U_1"   "PSA_U_2"   "PSA_U_3"
## [13] "PSA_U_4"   "PSA_U_5"   "PSA_U_7"   "PSA_U_8"   "PSA_U_10"  "PSA_U_11"
## [19] "PSA_L_1"   "PSA_L_2"   "PSA_L_3"   "PSA_L_4"   "PSA_L_5"   "PSA_L_7"
## [25] "PSA_L_8"   "PSA_L_10"  "PSA_L_11"
```

```
print(colnames(cts))
```

```
## [1] "HC_6"      "HC_7"      "HC_8"      "HC_9"      "HC_11"     "HC_12"
## [7] "HC_13"     "HC_14"     "HC_15"     "PSA_U_1"   "PSA_U_2"   "PSA_U_3"
## [13] "PSA_U_4"   "PSA_U_5"   "PSA_U_7"   "PSA_U_8"   "PSA_U_10"  "PSA_U_11"
## [19] "PSA_L_1"   "PSA_L_2"   "PSA_L_3"   "PSA_L_4"   "PSA_L_5"   "PSA_L_7"
## [25] "PSA_L_8"   "PSA_L_10"  "PSA_L_11"
```

```
#Perform DESeq
```

```
dds <- DESeqDataSetFromMatrix(cts, anot, ~ Type)
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

```
## Note: levels of factors in the design contain characters other than
## letters, numbers, '_' and '.'. It is recommended (but not required) to use
## only letters, numbers, and delimiters '_' or '.', as these are safe characters
## for column names in R. [This is a message, not a warning or an error]
```

```
dds$Type <- relevel(dds$Type, ref = "Healthy control skin")
```

```
#Pre-filter
```

```
keep <- rowSums(counts(dds) >= 10) >= 2
```

```
dds <- dds[keep, ]
```

```
# Run DESeq() and results()
```

```
dds_out <- DESeq(dds)
```

```
## estimating size factors
```

```
## Note: levels of factors in the design contain characters other than
## letters, numbers, '_' and '.'. It is recommended (but not required) to use
## only letters, numbers, and delimiters '_' or '.', as these are safe characters
## for column names in R. [This is a message, not a warning or an error]
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## Note: levels of factors in the design contain characters other than
## letters, numbers, '_' and '.'. It is recommended (but not required) to use
## only letters, numbers, and delimiters '_' or '.', as these are safe characters
## for column names in R. [This is a message, not a warning or an error]
```

```
## final dispersion estimates

## fitting model and testing

## Note: levels of factors in the design contain characters other than
## letters, numbers, '_' and '.'. It is recommended (but not required) to use
## only letters, numbers, and delimiters '_' or '.', as these are safe characters
## for column names in R. [This is a message, not a warning or an error]

## -- replacing outliers and refitting for 61 genes
## -- DESeq argument 'minReplicatesForReplace' = 7
## -- original counts are preserved in counts(dds)
```

```
## estimating dispersions
```

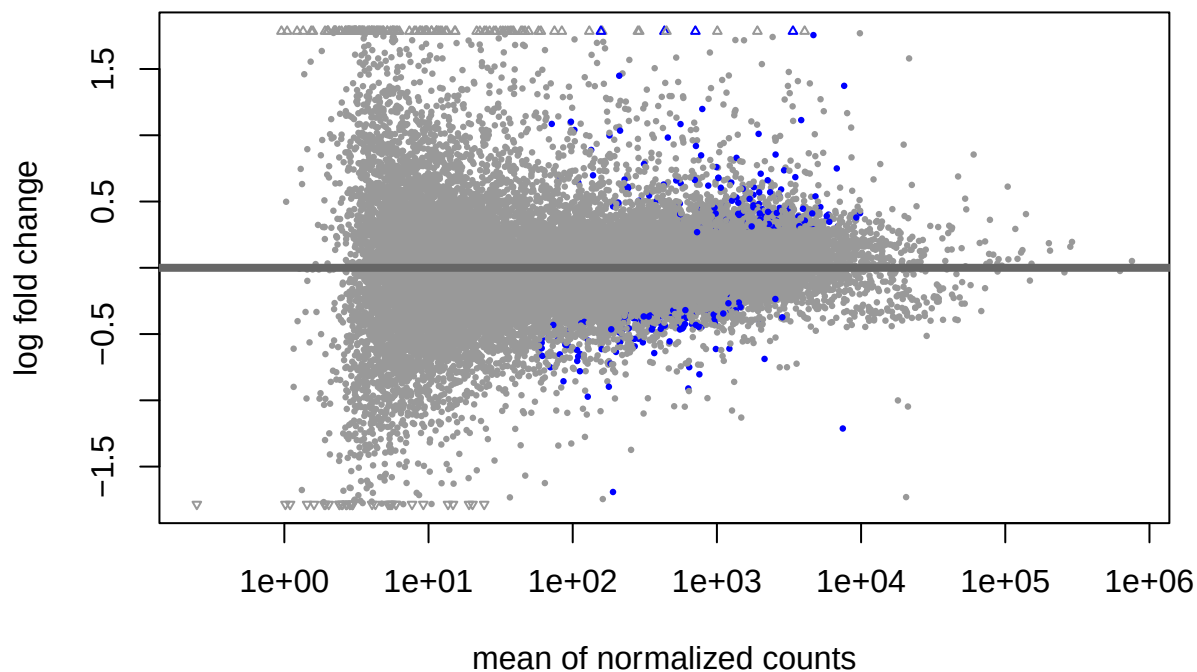
```
## fitting model and testing
```

```
## Note: levels of factors in the design contain characters other than
## letters, numbers, '_' and '.'. It is recommended (but not required) to use
## only letters, numbers, and delimiters '_' or '.', as these are safe characters
## for column names in R. [This is a message, not a warning or an error]
```

```
res <- results(dds_out)
# res
res_df <- as.data.frame(res)
res_df <- res_df[order(res_df$padj), ]
head(res_df)
```

```
##           baseMean log2FoldChange      lfcSE      stat      pvalue
## ENSG00000118096  408.4227      -0.4204035  0.08890994 -4.728420 2.262741e-06
## ENSG00000064393 1367.6206       0.8296215  0.17332607  4.786479 1.697330e-06
## ENSG00000170385 2339.2571       0.5702976  0.12177451  4.683226 2.823947e-06
## ENSG00000198363 3498.2890       0.6839619  0.14215663  4.811326 1.499322e-06
## ENSG00000198759  156.8427       3.0396421  0.66087977  4.599387 4.237352e-06
## ENSG00000167272  469.8044      -0.5553388  0.12069588 -4.601141 4.201828e-06
##                padj
## ENSG00000118096 0.01009843
## ENSG00000064393 0.01009843
## ENSG00000170385 0.01009843
## ENSG00000198363 0.01009843
## ENSG00000198759 0.01010185
## ENSG00000167272 0.01010185
```

```
plotMA(res)
```



```
#adding column to label genes as up/Down regulated or 'NA' if pvalue is insignificant
res_df <- data.frame(res)
res_df$DEG <- "NA"
res_df$DEG[res_df$log2FoldChange > 1 & res_df$pvalue < 0.05] <- "Up-regulated"
res_df$DEG[res_df$log2FoldChange < -1 & res_df$pvalue < 0.05] <- "Down-regulated"
head(res_df)
```

```
##          baseMean log2FoldChange      lfcSE      stat      pvalue
## ENSG00000282221    22.42173      0.16050989 0.26479685  0.6061624 0.54440694
## ENSG00000111671   224.09150     -0.23689863 0.16594946 -1.4275349 0.15342578
## ENSG00000110514  1805.58362      0.05969877 0.09104947  0.6556740 0.51203393
## ENSG00000086015   995.02117     -0.22622302 0.10079334 -2.2444244 0.02480511
## ENSG00000070770  5036.39555      0.04495675 0.16003706  0.2809146 0.77877591
## ENSG00000169740   345.60705     -0.20571091 0.13421008 -1.5327531 0.12533671
##          padj  DEG
## ENSG00000282221    NA  NA
## ENSG00000111671 0.4664577  NA
## ENSG00000110514 0.7712649  NA
## ENSG00000086015 0.2388467  NA
## ENSG00000070770 0.9124993  NA
## ENSG00000169740 0.4315023  NA
```

```
table(res_df$DEG)
```

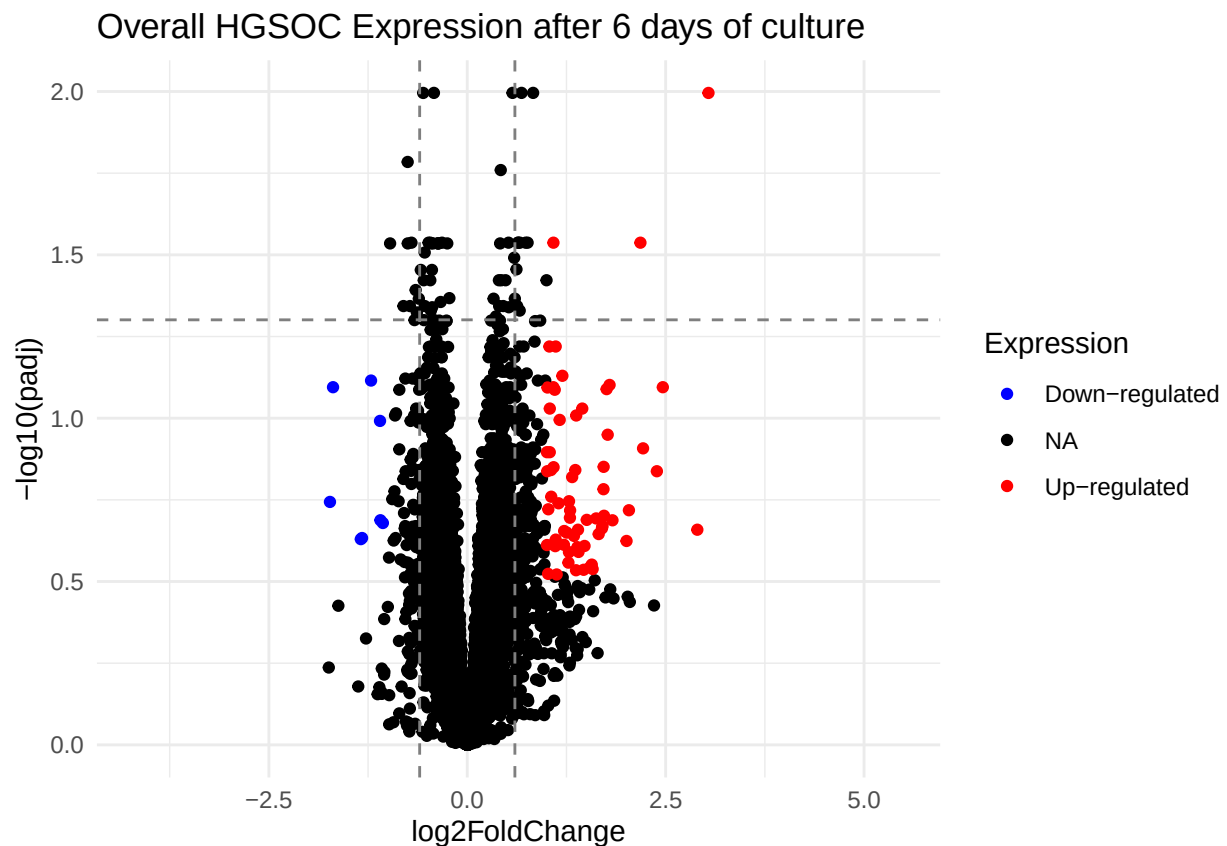
```
##
```



```
## Down-regulated      NA    Up-regulated
##           188      22870      305
```

```
ggplot(data = res_df, aes(x = log2FoldChange, y = -log10(padj), col = DEG)) +
  geom_point() +
  theme_minimal() +
  labs(
    title = "Overall HGSOC Expression after 6 days of culture",
    colour = "Expression"
  ) +
  scale_color_manual(values = c("blue", "black", "red")) +
  geom_vline(xintercept = c(-0.6, 0.6), col = "grey50", linetype = 'dashed') +
  geom_hline(yintercept = -log10(0.05), col = "grey50", linetype = 'dashed')
```

```
## Warning: Removed 9059 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
vsd <- vst(dds, blind = FALSE)
```

```
## Note: levels of factors in the design contain characters other than
## letters, numbers, '_' and '.'. It is recommended (but not required) to use
## only letters, numbers, and delimiters '_' or '.', as these are safe characters
## for column names in R. [This is a message, not a warning or an error]
```

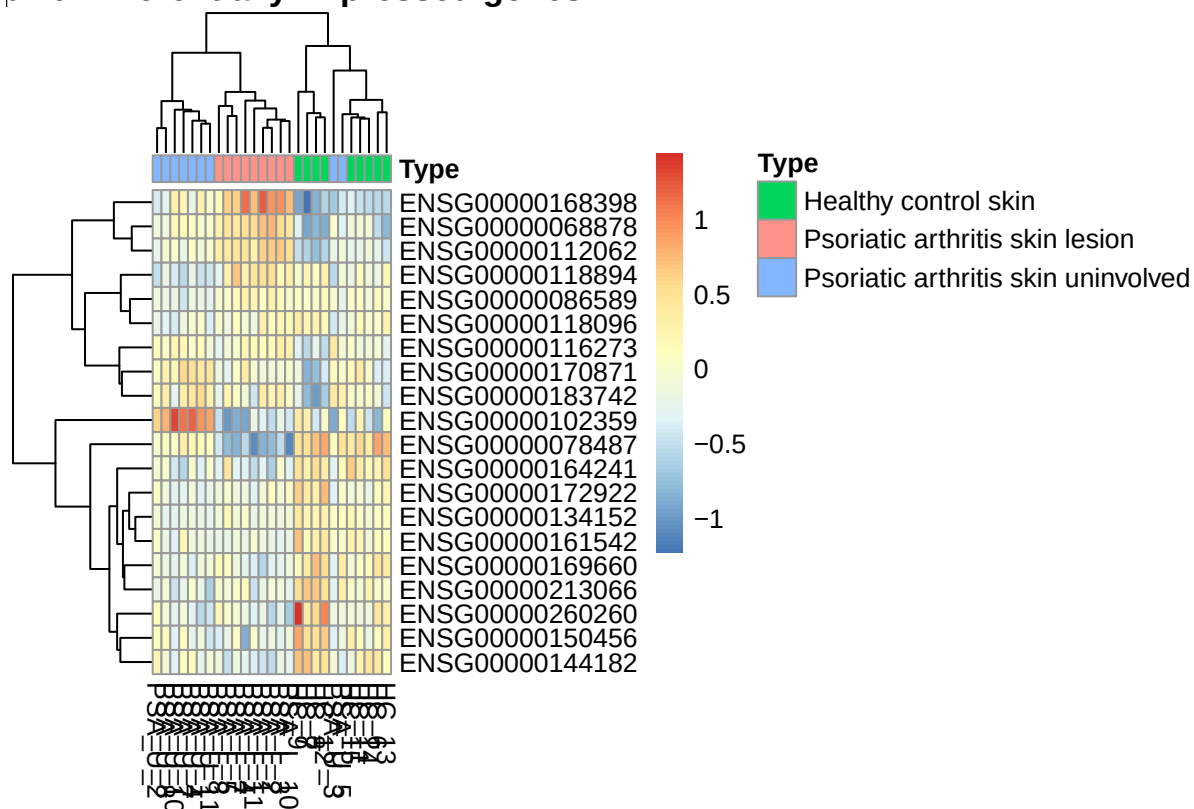
```
top_genes <- rownames(subset(res_df, padj < 0.05))[1:20]
mat <- assay(vsd)[top_genes, ]
mat <- mat - rowMeans(mat)

print(top_genes)
```

```
## [1] "ENSG00000078487" "ENSG00000260260" "ENSG00000169660" "ENSG00000168398"
## [5] "ENSG00000118894" "ENSG00000170871" "ENSG00000134152" "ENSG00000161542"
## [9] "ENSG00000183742" "ENSG00000086589" "ENSG00000213066" "ENSG00000150456"
## [13] "ENSG00000172922" "ENSG00000102359" "ENSG00000116273" "ENSG00000068878"
## [17] "ENSG00000164241" "ENSG00000112062" "ENSG00000144182" "ENSG00000118096"
```

```
pheatmap(mat,
  annotation_col = anot[, "Type", drop = FALSE],
  main = "Top 20 Differentially Expressed genes")
```

Top 20 Differentially Expressed genes



```
top_genes <- res_df %>%
  slice_min(padj, n=10)
top_genes <- head(top_genes, n=10)
# for table formatting
gene_ids <- sub("\\..*", "", rownames(top_genes)) # remove ensemble id version number
ensembl_ids <- AnnotationDbi::select(org.Hs.eg.db,
  keys = gene_ids,
  column = c("SYMBOL", "GENENAME"),
  keytype = "ENSEMBL")
```

ENSEMBL	SYMBOL	GENENAME	DEG
ENSG00000118096	IFT46	intraflagellar transport 46	NA
ENSG00000064393	HIPK2	homeodomain interacting protein kinase 2	NA
ENSG00000170385	SLC30A1	solute carrier family 30 member 1	NA
ENSG00000198363	ASPH	aspartate beta-hydroxylase	NA
ENSG00000198759	EGFL6	EGF like domain multiple 6	Up-regulated
ENSG00000167272	POP5	POP5 homolog, ribonuclease P/MRP subunit	NA
ENSG00000232977	LINC00327	long intergenic non-protein coding RNA 327	NA
ENSG00000186591	UBE2H	ubiquitin conjugating enzyme E2 H	NA
ENSG00000078487	ZCWPW1	zinc finger CW-type and PWWP domain containing 1	NA
ENSG00000134152	KATNBL1	katanin regulatory subunit B1 like 1	NA

'select()' returned 1:1 mapping between keys and columns

```
ensembl_ids %>%
  mutate(DEG = top_genes$DEG) %>%
  kbl() %>%
  kable_classic(full_width = F, html_font = "Cambria")
```

Performing PCA analysis

```
## Changing top_genes to include all genes used in the DEseq Differential expression
## analysis
norm_counts <- counts(dds, normalized = FALSE)
top_genes <- res_df %>%
  slice_min(padj, n=17240)
mat <- norm_counts[rownames(top_genes), ]
mat <- log2(mat + 1)
mat_scaled <- t(scale(t(mat)))
annotation_col <- data.frame(
  Type = colData(dds)$Type
)
rownames(annotation_col) <- colnames(mat_scaled)

# expression_matrix: genes x samples seq (RNA Seq File)
pca <- prcomp(t(mat_scaled), scale. = FALSE)
pca_df <- data.frame(
  PC1 = pca$x[,1],
  PC2 = pca$x[,2],
  Type = colData(dds)$Type
)
#
# ggplot(pca_df, aes(x = PC1, y = PC2, colour = row.names(pca_df))) +
#   geom_point(size = 3) +
#   theme_classic() +
#   labs(
#     x = paste0("PC1"),
#     y = paste0("PC2"),
#     title = "PCA of each RNA-seq Run",
```

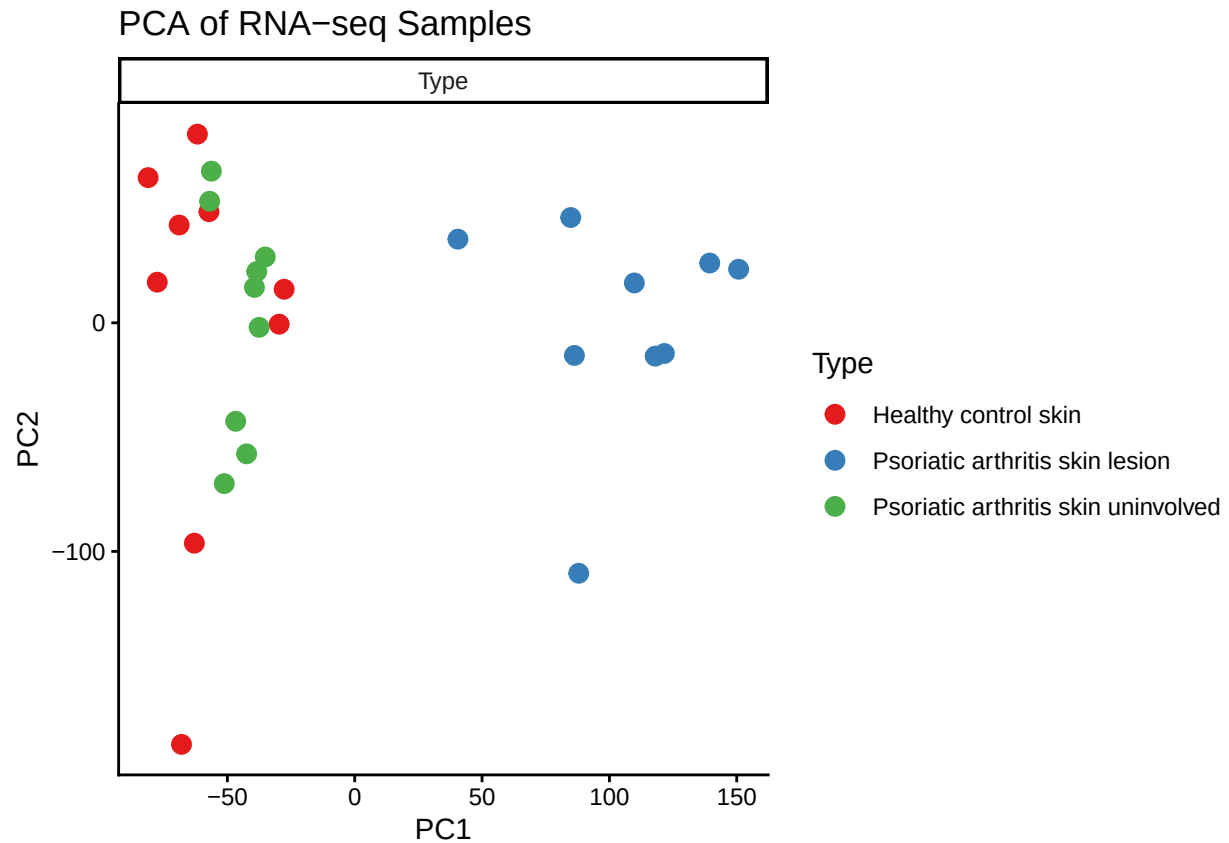
```

#   colour = "RNA-seq Run (patient_day)"
#   )+
#   scale_color_brewer(palette="Spectral")
#
# expression_matrix: splitting matrix based on patient and day (RNA Seq File)
pca_df2 <- pca_df %>%
  mutate(
    Type = as.character(Type),
  ) %>%
  pivot_longer(cols = c(Type),
               names_to = "ColorBy",
               values_to = "Group")

# # plot PC variance in a scree plot
# variance_pca = pca$sdev^2
# # proportion variance
# variance_pca / sum(variance_pca) -> prop_pca
#
# plot(prop_pca,
#       xlab = "Principal Components",
#       ylab = "Proportion of Variance Explained",
#       ylim = c(0, 1))

ggplot(pca_df2, aes(x = PC1, y = PC2, colour = Group)) +
  geom_point(size = 3) +
  theme_classic()+
  facet_wrap(~ ColorBy, scales = "free") +
  labs(
    title = "PCA of RNA-seq Samples",
    colour = "Type",
    x = paste0("PC1"),
    y = paste0("PC2"),
  )+
  scale_color_brewer(palette="Set1")

```



Short interpretation of the results

There is a significant difference in gene expression between normal and lesion samples, the top differential expressed genes are mostly up regulated as per the volcano plot.